Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/25/2014
PI Name:	Poirier, David Sc.D.		
Project Title:	Research for ESA Space Experiments on Directional Solidification of Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Tucson	**State:**	AZ
Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/15/2008	**End Date:**	06/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Expedition 19/20 NOTE: Extended to 6/30/2014 per NSSC information (Ed., 4/9/14) NOTE: Extended to 5/14/2014 per PI and NSSC information (Ed., 4/5/2013) NOTE: End date extended to 5/14/2013 (from 5/14/2011) in Dec 2010 per NSSC information (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	Professor Olaya Florez of the Universidad de Colombia was with the research team at The University of Arizona while he was on sabbatical for most of 2013.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX08AN49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this research program two experiments were processed on the International Space Station (ISS) to study the effect of microgravity on directionally solidified alloy microstructures and segregation. The original research included the two ISS-experiments, but a third was processed in late spring/early summer on the ISS. The third sample has yet to be returned to Earth for our analysis. Another proposal was submitted and funded to continue the work with the additional funds. Subsequent funding will be provided with the new grant. [NOTE (Ed., 8/25/2014): This project is being continued as a new grant--"Directional Solidification Experiments on the International Space Station," grant NNX14AN87G, with period of performance of 7/18/2014-1/17/2016.] We are studying directional solidification (DS) of alloys solidified in presence of 1-g (on Earth) and in 0-g (in space on the ISS). In this way we can compare the microstructures and macrostructures of samples solidified on Earth and in space to see the effects of convection. We have been carrying out the study by examining the DS-alloy and by computer simulations. Based on the research, it is clear that thermosolutal convection, which always occurs when alloys solidify on Earth, masks the effects of diffusional transport during solidification. With terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, directional solidification occurs with no thermosolutal convection, so the convection and attendant macrosegregation are almost completely eliminated, and the growth of the cast dendritic- microstructure is controlled by a diffusional processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to a process known as directional solidification, which is used for producing critical castings that are used in gas turbine engines. Gas turbine engines power aircraft (i.e., often called "jet engines") and are also used to generate electric power. These castings are made of vacuum melted alloys that are capable of withstanding the demanding conditions at high temperatures and consequently the metallurgical structure of the alloys must be of the highest quality. The research is leading to an improved understanding of how microstructures and segregation form in directionally solidified castings; hence the science of producing such castings will be enhanced as a result of the research. In addition, the research findings relate to many other types of castings, including billets and ingots, of alloys (i.e., steel, aluminum alloys, etc.) that are important in the "manufacturing chain."
Task Progress:	In this research program two experiments were processed on the International Space Station (ISS) to study the effect of microgravity on directionally solidified alloy microstructures and segregation. The original research included the two ISS-experiments, but a third was processed in late spring/early summer on the ISS. The third sample has yet to be returned to Earth for our analysis. Another proposal was submitted and funded to continue the work with the additional funds. Subsequent funding will be provided with the new grant. At Cleveland State University (CSU) terrestrial directional solidification experiments on Al-7%Si alloys were done using different growth conditions to effect variations in dendritic metrics. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is progressing at the University of Arizona (UA). Based on previous research, it is clear that thermosolutal convection, which always occurs when alloys solidify on Earth, masks the effects of diffusional transport during solidification. With terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, directional solidification occurs with no thermosolutal convection, so the convection and attendant macrosegregation are almost completely eliminated, and the growth of the cast dendritic- microstructure is controlled by a diffusional processes. We can also compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. We have found, as hypothesized, that the dendritic metrics of the space-grown samples have dendritic characteristics compare very well to a model for growth in the absence of convection. Earth-grown samples do not compare well, which indicates that the convection in Earth grown alloys does indeed play a role in microstructural development during the solidification process. Additional evidence is that Earth-grown samples exhibit a type of macrosegregation called "steepling." We see the convection by computer simulations and predict and measure the steepling. The two samples grown aboard the ISS do not have steepling. The technological significance of the research is that it provides scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft and land-based power generators. The research is being done by the team at The University of Arizona and Cleveland State University. We are part of MICAST (Microstructure Formation in Casting of Technical Alloys) led by ESA-researchers in the German Aerospace Center in Germany. One sample was processed in the ISS in August, 2010 and a second was processed on the ISS in Feb. 20011. Both samples have been returned to Earth and have been under study at CSU and MSFC. Computer simulations have been ongoing at UA. As mentioned above, we await the third sample from the ISS for study. NOTE (Ed., 8/25/2014): This project is being continued as a new grant--"Directional Solidification Experiments on the International Space Station," grant NNX14AN87G, with period of performance of 7/18/2014-1/17/2016. See that task for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Poirier DR. "Density, viscosity, and diffusion coefficients in hypoeutectic Al-Si alloys: an assessment of available data." Metallurgical and Materials Transactions B. 2014 Aug;45(4):1345-54. http://dx.doi.org/10.1007/s11663-014-0037-8 , Aug-2014
Articles in Peer-reviewed Journals	Tewari SN, Grugel RN, Poirier DR. "An evaluation of primary trunk diameters in directionally solidified Al-Si alloys." Metallurgical and Materials Transactions A. 2014 Oct;45(11):4758-61. http://dx.doi.org/10.1007/s11661-014-2438-3 , Oct-2014

Articles in Peer-reviewed Journals	Ghods M, Johnson L, Lauer M, Grugel RN, Tewari SN, Poirier DR. "Radial macrosegregation and dendrite clustering in directionally solidified Al–7Si and Al–19Cu alloys." Journal of Crystal Growth. 2016 May;441:107-16. http://dx.doi.org/10.1016/j.jcrysgro.2016.02.014 , May-2016
Articles in Peer-reviewed Journals	Masoud G, Upadhyay SR, Rajamure RS, Tewari SN, Grugel RN, Poirier DR, Lauer M. "Primary dendrite array morphology in Al-7 wt% Si alloy samples directionally solidified aboard the International Space Station." Journal of Crystal Growth. 2021 May;562:126077. https://doi.org/10.1016/j.jcrysgro.2021.126077 , May-2021
Papers from Meeting Proceedings	Angart S (speaker), Lauer M, Tewari SN, Grugel RN, Poirier DR. "Comparison of Directionally Solidified Samples Solidified Terrestrially and Aboard the International Space Station." Presented at 61st Annual Technical Conference of the Investment Casting Institute, Covington, KY, October 5-8, 2014. Proceedings of the 61st Annual Technical Conference of the Investment Casting Institute, Covington, KY, October 5-8, 2014, paper 6 (23 pages). See also http://ntrs.nasa.gov/search.jsp?R=20140016721 , Oct-2014